A Simulation Study of Indirect Comparison Methods

Placement Project Report

September 4, 2025

Executive Summary

In this report, we present a simulation study comparing three indirect comparison methods: the Bucher method, Matching-Adjusted Indirect Comparison (MAIC), and Simulated Treatment Comparison (STC).

Key Findings:

- MAIC and STC exhibit greater bias and standard errors than the Bucher method when sample sizes are small.
- The Bucher method becomes increasingly biased as effect modifier strength increases.
- MAIC shows significantly higher bias and standard errors with low population overlap, whilst STC bias and standard errors increase by a far smaller extent.
- Both MAIC and STC require adjustment of all effect modifiers to minimise bias and achieve optimal accuracy.

Contents

1 Introduction

In the pharmaceutical industry, it is common for a company to have individual patient data (IPD) on its own clinical trial but only aggregate data for a competitor trial. Aggregate data typically consists of:

- Average treatment effects across patient populations (e.g. means, odd ratios, hazard ratios).
- Summary statistics for baseline patient characteristics, such as:
  - Mean and standard deviation for continuous characteristics (e.g. age or BMI).
  - Proportions for binary characteristics (e.g. sex).

Let us denote the company's trial as AB (comparing treatments A and B), and the competitor trial as AC (comparing treatments A and C). The company wishes to compare the effectiveness of treatments B and C, but lacks a clinical trial comparing the two directly. Instead, it can estimate the comparative effectiveness of treatments B and C using the AB and AC trials, a method known as an **indirect comparison** (Figure 1).

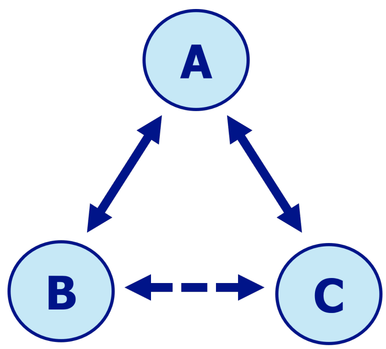

Figure 1: Diagram of an indirect comparison

The Bucher method is an indirect comparison method developed in 1997 (Bucher et al., 1997). However, due to limitations in the Bucher method, alternative indirect comparison methods known as **population-adjusted indirect comparisons** have been developed. Two notable population-adjusted indirect comparison methods are Matching-Adjusted Indirect Comparison (MAIC) and Simulated Treatment Comparison (STC).

In this report, we will:

- Present the motivation for population-adjusted indirect comparisons.
- Outline the mechanisms of the Bucher method, MAIC, and STC.
- Describe the design of a simulation study evaluating these methods.
- Analyse and interpret the study results.
- Draw conclusions about each method's strengths and limitations.

2 Motivation for Population-Adjusted Indirect Comparisons

To explain the motivation behind population-adjusted indirect comparisons, we first introduce the concept of an **effect modifier**. An effect modifier is a variable that influences the relationship between a treatment and an outcome. For instance, a treatment may be more effective in younger patients than older ones, making age an effect modifier.

The Bucher method assumes that the distributions of effect modifiers are the same in all populations P, which implies that relative treatment effects are also the same across these populations. This assumption is known as the **constancy of relative effects** (Phillippo et al., 2016). In practice however, the distributions of effect modifiers in the AB and AC trial population are likely to be different. These differences lead to bias, making it impossible to generate accurate estimates for the effectiveness of treatment B compared to treatment C.

As a result, population-adjusted indirect comparison methods such as MAIC (Signorovitch et al., 2010) and STC (Caro and Ishak, 2010) were developed in the early 2010s. These methods adjust for differences in effect modifier distributions in the AB and AC trial populations, allowing us to generate more accurate relative treatment effect estimates.

3 Indirect Comparison Methodologies

3.1 Bucher Methodology

We now outline the process for the Bucher method (Bucher et al., 1997; Phillippo et al., 2018).

Our goal is to compare the effectiveness of treatments B and C, which we assess by estimating the relative C vs. B treatment effect in a target population P, denoted by $d_{BC(P)}$.[1]

$d_{BC(P)}$ is calculated as the difference between the C vs. A and B vs. A relative treatment effects:

$$d_{BC(P)} = d_{AC(P)} - d_{AB(P)} \tag{1}$$

We define the true population-level expected outcomes for treatments A, B, and C within the AB trial population as $\mu_{A(AB)}$, $\mu_{B(AB)}$, and $\mu_{C(AB)}$, respectively. From the AB trial data, we estimate $\mu_{A(AB)}$ and $\mu_{B(AB)}$ using the corresponding outcome statistics $\overline{Y}_{A(AB)}$ and $\overline{Y}_{B(AB)}$. Similarly for the AC trial, we define parameters $\mu_{A(AC)}$, $\mu_{B(AC)}$, and $\mu_{C(AC)}$, and estimate $\mu_{A(AC)}$ and $\mu_{C(AC)}$ from the corresponding outcome statistics $\overline{Y}_{A(AC)}$ and $\overline{Y}_{C(AC)}$.

Next, we select an appropriate transformation scale depending on the outcome type, for instance binary outcomes are often modeled on the log-odds scale while continuous outcomes use the identity scale (Dias et al., 2013).

We estimate the relative treatment effects $d_{AB(AB)}$ and $d_{AC(AC)}$ in the AB and AC trial populations by constructing estimators $\hat{d}_{AB(AB)}$ and $\hat{d}_{AC(AC)}$, given by:

$$\hat{d}_{AB(AB)} = g(\overline{Y}_{B(AB)}) - g(\overline{Y}_{A(AB)}) \tag{2a}$$
$$\hat{d}_{AC(AC)} = g(\overline{Y}_{C(AC)}) - g(\overline{Y}_{A(AC)}) \tag{2b}$$

where $g(.)$ is the link function used to convert outcome data to the appropriate scale (Rothman et al., 2008). For instance:

- If a log scale is used, $g(y) = log(y)$ and
- If the identity scale is used, $g(y) = y$.

Under the constancy of relative effects assumption, the relative treatment effects are equal across all populations, meaning that:

$$d_{AB(AB)} = d_{AB(AC)} = d_{AB(P)}$$
$$d_{AC(AB)} = d_{AC(AC)} = d_{AC(P)}$$

Thus, $\hat{d}_{AB(AB)}$ and $\hat{d}_{AC(AC)}$ become unbiased estimators for $d_{AB(P)}$ and $d_{AC(P)}$. Consequently, we estimate $d_{BC(P)}$ by substituting $\hat{d}_{AB(AB)}$ and $\hat{d}_{AC(AC)}$ into (1), giving:

$$\hat{d}_{BC(P)} = \hat{d}_{AC(AC)} - \hat{d}_{AB(AB)} \tag{3}$$

where $\hat{d}_{BC(P)}$ is the estimator for $d_{BC(P)}$.

[1]In the notation d_{BC}, the first letter (B) denotes the reference or baseline treatment, and the second letter (C) denotes the comparison or intervention treatment. By convention in comparative effectiveness research, d_{BC} represents the relative effect of treatment C versus treatment B, and is calculated as:

$$d_{BC} = \text{effect of treatment } C - \text{effect of treatment } B$$

Thus a positive value of d_{BC} is obtained when treatment C performs better than treatment B.

3.2 MAIC Methodology

Let us now introduce the population-adjusted indirect comparison method MAIC.

The core concept of MAIC is to re-weight the AB trial IPD so that the distribution of effect modifiers in the re-weighted AB trial population matches the distribution in the AC trial population (Jiang et al., 2024; Remiro-Azócar et al., 2021; Signorovitch et al., 2013; Weber et al., 2020).

Let us denote for each i-th, $(i = 1, \ldots, N)$ patient in the AB trial:

- A covariate vector of K baseline characteristics $\mathbf{X}_{i(AB)} = (X_{i,1(AB)}, \ldots, X_{i,K(AB)})$, e.g. age, sex, BMI.
- A treatment indicator $t_{i(AB)} \in \{A, B\}$.
- An observed outcome $Y_{i(AB)}$.

From the AC trial we have:

- A vector $\overline{\boldsymbol{X}}_{(AC)} = (\overline{X}_{1(AC)}, \ldots, \overline{X}_{K(AC)})$ of K summary statistics for the baseline characteristics.
- An estimate $\hat{d}_{AC(AC)}$ of the C vs. A treatment effect in the AC trial population.

We denote the covariate vector of effect modifiers per patient i by $\boldsymbol{X}^{EM}_{i(AB)} \subset \boldsymbol{X}_{i(AB)}$, and the vector of effect modifier summary statistics $\overline{\boldsymbol{X}}^{EM}_{(AC)} \subset \overline{\boldsymbol{X}}_{(AC)}$.

A single weight w_i is calculated for each patient in the AB trial, such that the weighted mean of each effect modifier in the AB trial matches the mean of the effect modifier in the AC trial, i.e.

$$\frac{\sum_{i=1}^{N} w_i \boldsymbol{X}^{EM}_{i(AB)}}{\sum_{i=1}^{N} w_i} = \overline{\boldsymbol{X}}^{EM}_{(AC)}$$

Weights are based on how closely each patient's effect modifier values align with the AC trial's summary statistics; patients whose characteristics more closely match the AC trial population receive higher weights.

The weights are estimated using a propensity score logistic regression model:

$$ln(w_i) = \alpha_0 + \boldsymbol{X}^{EM}_{i(AB)} \boldsymbol{\alpha_1}$$

where α_0 and $\boldsymbol{\alpha_1}$ are the regression parameters. The regression parameters are estimated by minimising the objective function:

$$Q(\boldsymbol{\alpha_1}) = \sum_{i=1}^{N} \exp\left(\boldsymbol{X}^{EM}_{i(AB)} \boldsymbol{\alpha_1}\right) \quad \text{when} \quad \overline{\boldsymbol{X}}^{EM}_{(AC)} = 0$$

using the BFGS algorithm, an iterative optimisation method (Nocedal and Wright, 2006). The estimated weight $\hat{w}_i$ for each patient i is found by $\hat{w}_i = \exp(\boldsymbol{X}^{EM}_{i(AB)} \hat{\boldsymbol{\alpha}}_{\mathbf{1}})$.

We then estimate the outcomes $\mu_{A(AC)}$ and $\mu_{B(AC)}$ for treatments A and B in the AC population by calculating weighted averages of the observed outcomes $Y_{i(AB)}$:

$$\overline{Y}_{A(AC)} = \frac{\sum_{i \in \mathcal{A}} \hat{w}_i Y_{i(AB)}}{\sum_{i \in \mathcal{A}} \hat{w}_i}$$
$$\overline{Y}_{B(AC)} = \frac{\sum_{i \in \mathcal{B}} \hat{w}_i Y_{i(AB)}}{\sum_{i \in \mathcal{B}} \hat{w}_i}$$

where $\mathcal{A} = \{i : t_{i(AB)} = A\}$ and $\mathcal{B} = \{i : t_{i(AB)} = B\}$ are the sets of patients in the AB trial who received treatments A and B, respectively.

The indirect comparison can then be carried out using the following formulae:

$$\hat{d}_{AB(AC)} = g(\overline{Y}_{B(AC)}) - g(\overline{Y}_{A(AC)}) \tag{4}$$

$$\hat{d}_{BC(AC)} = \hat{d}_{AC(AC)} - \hat{d}_{AB(AC)} \tag{5}$$

We have estimated the C vs. B treatment effect in the AC trial population. We now introduce the **shared effect modifier assumption** (Phillippo et al., 2016). This states that effect modifiers operate in the same manner across all populations. More specifically, this means that for our set of treatments, the same set of effect modifiers exist in all populations, and the change in treatment effects caused by each effect modifier is the same in each population. This allows us to project the relative C vs. B treatment effect in the AC trial population onto any population P, such that:

$$\hat{d}_{BC(P)} = \hat{d}_{BC(AC)} \tag{6}$$

3.3 STC Methodology

We now describe an alternative population-adjusted indirect comparison method STC.

STC uses outcome regression rather than re-weighting to adjust the AB trial population (Foch et al., 2022; Remiro-Azócar et al., 2021; Phillippo et al., 2018). The core idea is to fit a regression model to the AB trial IPD to capture how effect modifiers influence the treatment effect. Then, we substitute the AC trial effect modifier summary statistics into the regression model. This predicts the outcomes that would be expected in the AC trial population.

We define the outcome regression model by:

$$g(\eta_{i(AB)}) = \beta_0 + (\beta_B + \boldsymbol{\beta}_1^T \boldsymbol{X}_{i(AB)}^{EM}) \cdot \mathbb{I}(t_{i(AB)} = B)$$

where:

- $\eta_{i(AB)}(\boldsymbol{X_{i(AB)}})$ is the expected outcome of a patient i with covariate values $\boldsymbol{X}_{i(AB)}$.
- $t_{i(AB)} \in \{A, B\}$ is the treatment indicator.
- $g(.)$ is the appropriate link function.
- β_0 is the model intercept, representing the expected outcome for treatment A before accounting for treatment effects or effect modifiers.
- β_B is the baseline relative B vs. A treatment effect.
- $\boldsymbol{\beta}_1$ is a vector of effect modifier coefficients quantifying how each effect modifier in $\boldsymbol{X}_{i(AB)}^{EM} \subset \boldsymbol{X}_{i(AB)}$ modifies the treatment effect.
- $\mathbb{I}(.)$ is the indicator function.

$\overline{\boldsymbol{Y}}_{A(AC)}$ and $\overline{\boldsymbol{Y}}_{B(AC)}$ can then be predicted from the outcome regression as follows:

$$\overline{\boldsymbol{Y}}_{A(AC)} = g^{-1}(\hat{\beta}_0)$$

$$\overline{\boldsymbol{Y}}_{B(AC)} = g^{-1}(\hat{\beta}_0 + \hat{\beta}_B + \hat{\boldsymbol{\beta}}_1^T \overline{\boldsymbol{X}}_{(AC)}^{EM})$$

where $\overline{\boldsymbol{X}}_{(AC)}^{EM}$ represents the effect modifier summary statistics, and $\hat{\beta}_0$, $\hat{\beta}_B$, and $\hat{\boldsymbol{\beta}}_1$ are observed estimates for β_0, β_B, and $\boldsymbol{\beta}_1$.

As in MAIC, we assume that the shared effect modifier assumption holds. We then proceed with the indirect comparison as in MAIC, using equations (4), (5), and (6).

4 Simulation Study Design

We conduct a simulation study to compare the performance of the Bucher method, MAIC, and STC across four scenario groups (Morris et al., 2019; Phillippo et al., 2020).

4.1 Data Generation

For each simulation iteration, we generate two datasets:

- An AB trial of individual patient data (IPD).
- An AC trial from which we calculate summary statistics.

Each trial includes N participants, with equal allocation to treatments.

4.1.1 Model Specification

We generate binary outcomes in both trials using logistic regression models incorporating two continuous effect modifiers:

***AB* Trial:**

$$\text{logit}(p_{it}) = b_0 + \Big[b_{\text{trt}} + m_X(X_{it(AB)} - \mu_{X(AB)}) + m_{X'}(X'_{it(AB)} - \mu_{X'(AB)})\Big] \cdot \mathbb{I}(t = B)$$

***AC* Trial:**

$$\text{logit}(p_{it}) = c_0 + \Big[c_{\text{trt}} + m_X(X_{it(AC)} - \mu_{X(AB)}) + m_{X'}(X'_{it(AC)} - \mu_{X'(AB)})\Big] \cdot \mathbb{I}(t = C)$$

where:

- p_{it} represents the probability of the outcome for patient i on treatment t.
- X and X' are effect modifier variables.
- $\mu_{X(AB)}$ and $\mu_{X'(AB)}$ are the effect modifier means in the AB trial population, which we use to centre the effect modifier variables in both trial populations. This improves both interpretability and validity of treatment effect estimates.
- b_0 and c_0 represent the baseline log-odds of the outcome for patients receiving treatment A in the AB and AC trial populations.
- b_{trt} and c_{trt} are the treatment effects associated with treatments B and C respectively.
- m_X and $m_{X'}$ are coefficients indicating the **effect modifier strengths** of X and X' in both the AB and AC trial populations. We define effect modifier strength as how strongly an effect modifier variable modifies the effect of a treatment on an outcome. The greater the effect modifier strength, the more sensitive treatment effects are to differences in the distribution of effect modifiers.

4.1.2 Parameter Settings

We set the parameters in the above outcome models as follows:

- Effect modifier distributions in the AB trial population follow normal distributions $X_{(AB)} \sim N(1, 0.5^2)$ and $X'_{(AB)} \sim N(0.5, 0.1^2)$ (Figure 2). The standard deviations 0.5 and 0.1 are denoted by $\sigma_{X(AB)}$ and $\sigma_{X'(AB)}$ respectively.
- Effect modifier distributions in the AC trial population follow normal distributions, with mean and standard deviation parameters varying according to the degree of overlap with the AB trial population. This will be further detailed in Section 4.2, where we explain the investigated scenario groups.
- Baseline log-odds of $b_0 = 1$ and $c_0 = 1.5$.

- Treatment effects of $b_{trt} = -2$ and $c_{trt} = -1.5$.

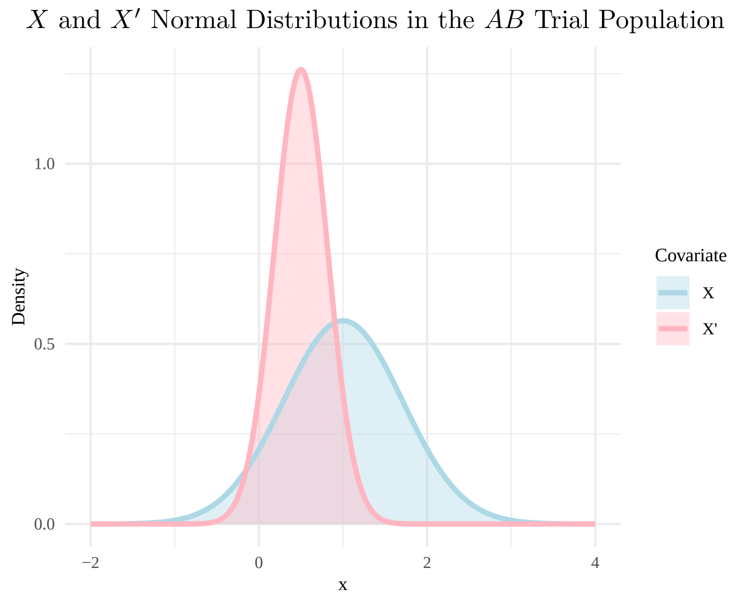

Figure 2: Diagram of X and X' normal distributions in the AB trial population

4.2 Investigated Scenarios

We vary four key factors to evaluate the Bucher method, MAIC, and STC across different scenarios.

Sample Size

We test three sample sizes of $N = 100$, $N = 500$, and $N = 1000$.

Effect Modifier Strength

We vary the effect modifier strength through a parameter EMS, in which we define the effect modifier coefficients m_X and $m_{X'}$ to be:

- $m_X = EMS \times \sigma_{X(AB)}$.
- $m_{X'} = EMS \times \sigma_{X'(AB)}$.

We investigate EMS values of 0.1, 0.5, and 1.

Population Overlap

We alter the degree of overlap between the AB and AC trial populations by defining the X and X' effect modifier distributions in the AC trial as:

- $X_{(AC)} \sim N((1.1 + (1 - POL)^2)\,\mu_{X(AB)}, (0.75\,\sigma_{X(AB)})^2)$.
- $X'_{(AC)} \sim N((1.1 + (1 - POL)^2)\,\mu_{X'(AB)}, (0.75\,\sigma_{X'(AB)})^2)$.

where variable POL controls the degree of overlap, and investigate POL values of 1, 0.5, and 0.25 (Figure 3).

Different AB and AC Trial Population Overlaps (POL)

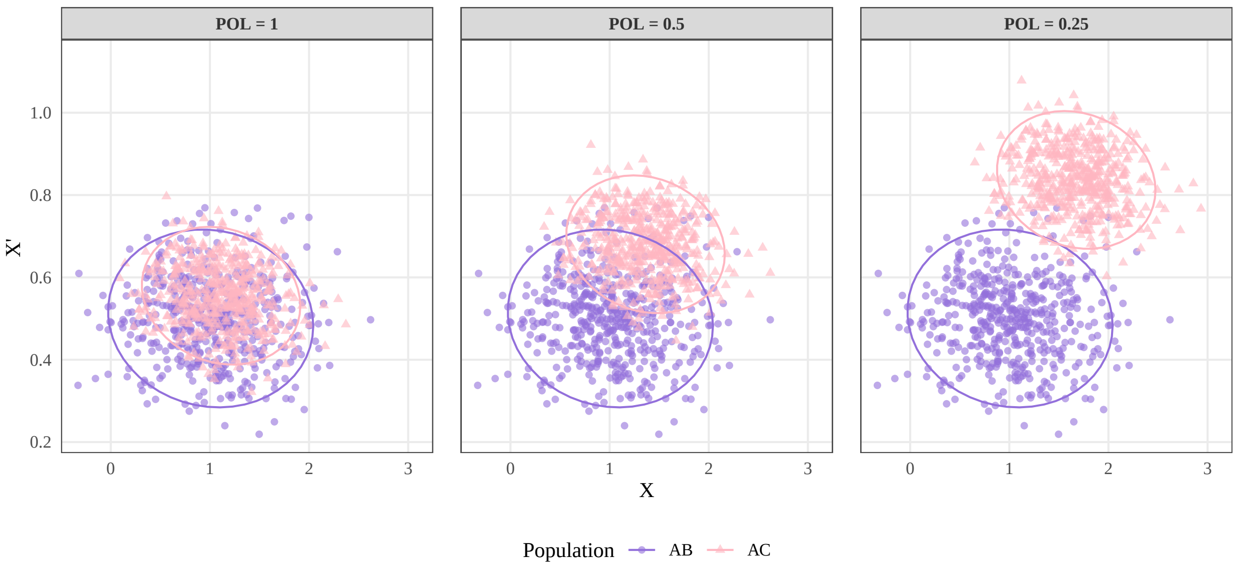

Figure 3: Scatterplot of $POL = 1$, 0.5, and 0.25

Effect Modifier Adjustment

We evaluate two adjustment approaches for MAIC and STC:

- Full adjustment, where both effect modifiers (X and X') are adjusted for in the model.
- Partial adjustment, where only X' is adjusted for.

The Bucher method does not adjust for effect modifiers, so the calculation is the same in both instances.

4.3 Simulation Procedure

We conduct 1,000 simulation iterations for each scenario. For each iteration we:

- Generate data based on the desired scenario parameters. When parameters are not being deliberately varied for our experiment, we choose the default parameters $N = 500$, $EMS = 0.5$, $POL = 0.75$, and adjustment for both effect modifiers.
- Apply the Bucher method, MAIC, and STC.
- Obtain and record for each method the following values:
  - Estimate of the C vs. B treatment effect.
  - Standard error of the C vs. B treatment effect estimate.
  - Confidence interval of the C vs. B treatment effect estimate.

4.4 Performance Measures

4.4.1 Non-Method-Specific Performance Measures

Following the simulation phase, we use the collected results to calculate the following performance metrics for all three methods.

Bias

Bias measures how far the estimated treatment effect is from the true value. It is calculated as the average difference between the estimated values and the true value across all simulation iterations. The closer the bias is to zero, the less biased the estimator.

$$\text{Bias} = \frac{1}{n}\sum_{j=1}^{n}(\hat{\theta}_j - \theta_{true})$$

where:

- θ_j is the estimate of the C vs. B treatment effect for the *j-th* simulation iteration.
- θ_{true} is the true C vs. B treatment effect value.
- n is the number of simulation iterations.

Empirical Standard Error

Empirical standard error represents the observed variability or precision of the estimator across simulation iterations. Lower empirical standard errors indicate more precise estimates. It is calculated as the standard deviation of the estimates:

$$\text{Empirical SE} = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(\hat{\theta}_j - \overline{\theta})^2}$$

Model Standard Error

Model standard error measures how well a statistical method assesses its own uncertainty. It is calculated as the average of standard errors reported by the method itself across all iterations of the simulation.

$$\text{Model SE}(\hat{\theta}) = \sqrt{\frac{1}{n}\sum_{j=1}^{n} SE(\hat{\theta}_j)^2}$$

where $SE(\hat{\theta}_j)$ is the standard error of the C vs. B treatment effect estimate for the *j-th* simulation iteration.

Ideally, the model standard error should closely match the empirical standard error (the actual observed variability in estimates) - this indicates that the model is accurately quantifying its uncertainty.

Coverage

Coverage is the proportion of simulation iterations where the true parameter value is within the confidence intervals.

$$\text{Coverage} = \frac{1}{n}\sum_{j=1}^{n}\mathbb{I}(\theta_{true} \in [L_j, U_j])$$

where $[L_j, U_j]$ is the confidence interval for the *j-th* simulation iteration.

For a 95% confidence interval, we expect approximately 0.95 coverage.

Monte Carlo Standard Errors

For each performance measure, we compute monte carlo standard errors to quantify the statistical uncertainty in our estimates due to the finite number of simulation iterations. Smaller monte carlo standard errors indicate more precise estimates. By monitoring the size of these errors, we can assess whether we have run enough simulation iterations to achieve our desired level of precision.

4.4.2 MAIC-Specific Performance Measures

In each MAIC iteration, statistics are calculated from the (rescaled[2]) weights, then averaged across iterations.

- **ESS**: the effective sample size.
- **ESS/N(%)**: the percentage of ESS to N.
- **Min, Q1, Median, Q3, Max**.
- **Mean**: (always 1 for rescaled weights).
- **Number of 0 weights**: (defined as < 0.001).
- **Number of $>$ 3 weights**.

ESS represents the effective number of patients remaining after weighting adjustments, given by:

$$ESS = \frac{(\sum_{i=1}^{N} w_i)^2}{\sum_{i=1}^{N} w_i^2}$$

where:

- w_i represents the weight for the i-th patient in the trial population.
- N represents the number of patients in the trial population.

ESS decreases as weights become more extreme, and a low ESS indicates that a small number of patients are significantly influencing the results (Phillippo et al., 2019). The number of near-zero weights and large weights also impact the ESS calculation - zero weights remove patients from the analysis while large weights give patients disproportionate influence. These statistics describe the distribution of the obtained weights, helping to evaluate the stability and validity of the MAIC.

4.5 Visualisation

We visualise results using dot plots for bias and empirical and model standard errors, centile plots for confidence intervals, and summary tables for the calculated statistics.

We will include the most notable figures and tables in our results section, but all figures and tables can be found in the appendices.

4.6 Package Workflow

The simulation study code is structured as an R package (Wickham and Bryan, 2023). We illustrate the package workflow in Figure 4, and describe the purpose of each function as follows:

- `gen_data()`: generates a set of IPD and aggregate data for the AB and AC trials, respectively.
- `bucher()`, `maic()`, and `stc()`: perform the Bucher method, MAIC, and STC.
- `calc_true()`: calculates the true C vs. B treatment effect.

[2]Rescaled weights are calculated as:

$$\tilde{w}_i = \frac{w_i}{\sum_{k=1}^{N} w_k} \cdot N$$

where:

- $\tilde{w}_i$ is the rescaled weight for patient i.
- w_i is the original weight for patient i.
- N is the number of patients in the trial population.

- `do_simulation()`: a wrapper function for the above functions, which facilitates the execution of the desired number of simulation iterations.
- `analyse_simulations()`: calculates performance metrics.
- `plot_bias()`, `plot_se()`, and `plot_coverage()`: draw plots for the relevant performance metrics.
- `format_table()`: creates tables summarising performance metrics.

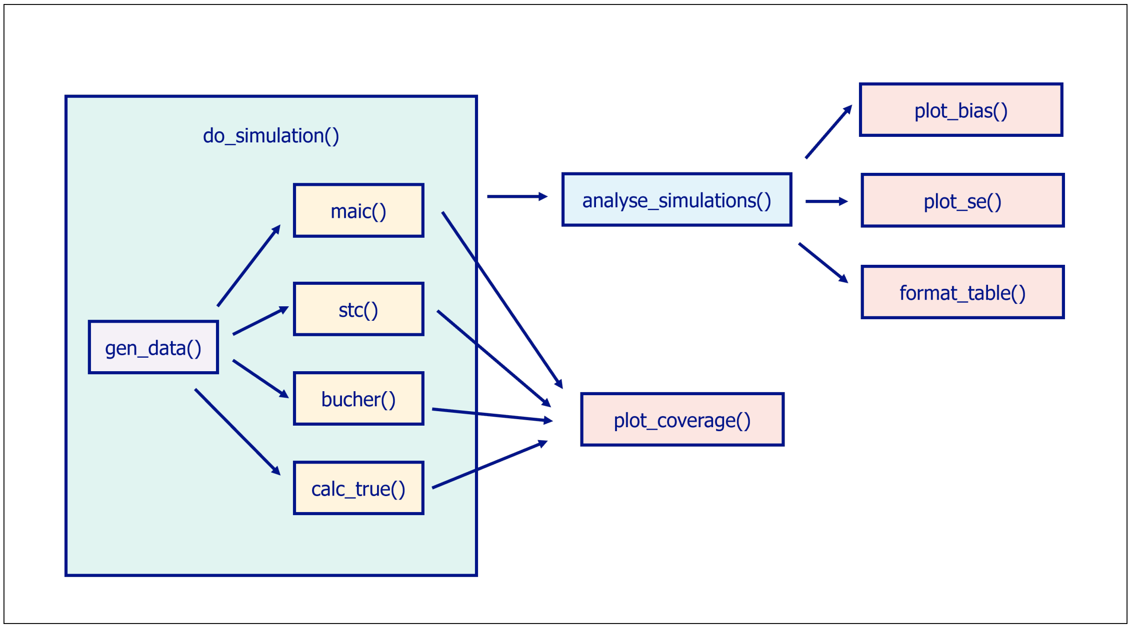

Figure 4: Diagram of Package Workflow

5 Results

We now discuss the results obtained from the simulation study.

With the exception of scenarios involving MAIC (and the Bucher method in the case of coverage) when $POL = 0.25$, model standard errors aligned well with empirical standard errors, coverage was close to 0.95, and monte carlo standard errors were low (less than 0.03). These results indicate that, in most scenarios, standard errors were well estimated, uncertainty was appropriately quantified, and 1000 simulations were sufficient to obtain reliable estimates of bias, standard errors, and coverage.

5.1 Sample Size

Sample sizes of 100, 500, and 1000 were simulated for all three methods.

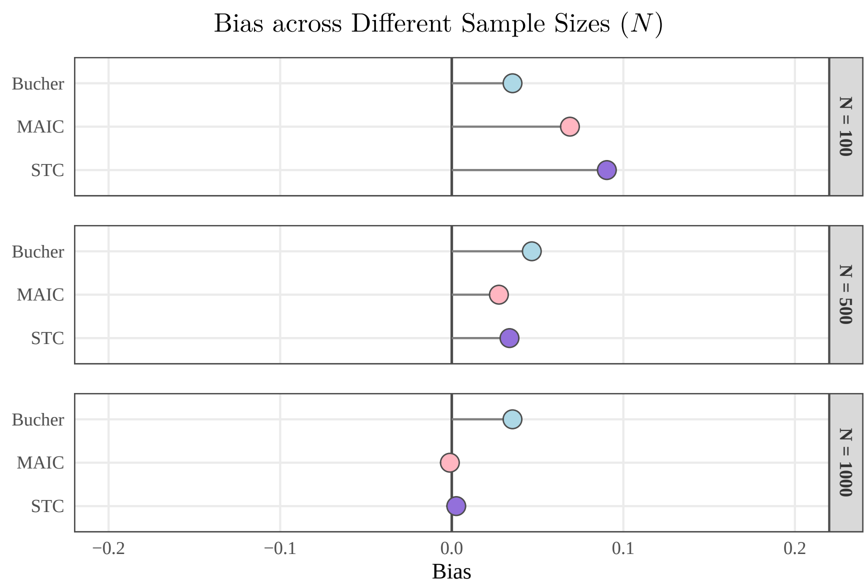

Figure 5: Bias for $N = 100$, 500, and 1000

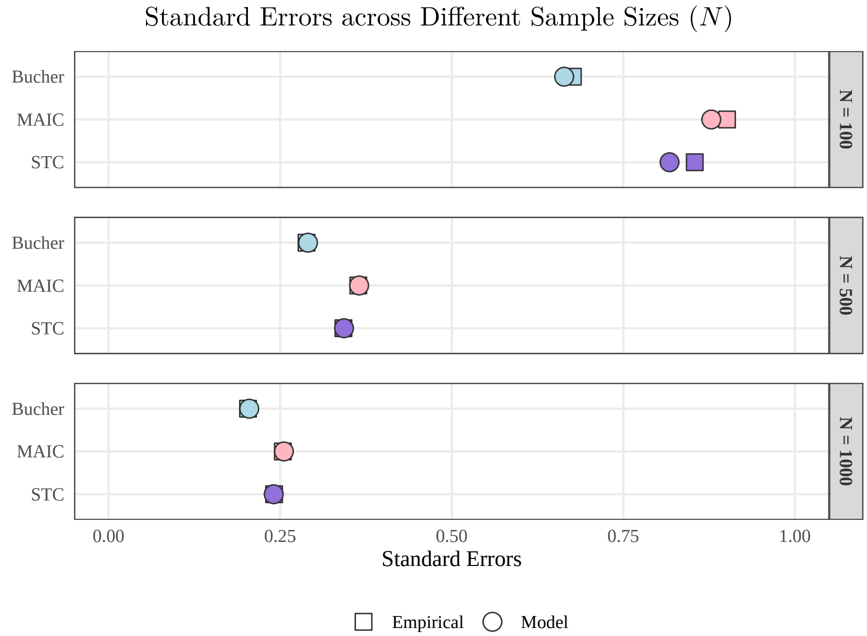

Figure 6: Empirical and Model SEs for $N = 100$, 500, and 1000

For a sample size N of 100, MAIC and STC showed larger biases than the Bucher method (Figure 5). However, as N increased to 500 and 1000, MAIC and STC both showed smaller biases than Bucher.

There could be several reasons for the poor performance of MAIC and STC when N is small. STC is based on a logistic regression model which inherently suffers from small-sample bias, also known as finite sample bias. With small sample sizes STC may also be more susceptible to overfitting - when the model fits to random noise rather than true signals. For MAIC, the estimation of weights can become highly unstable with small sample sizes, as with fewer patients very large weights are often assigned to a small number of individuals.

When N was 100, empirical standard errors for MAIC and STC were greater than for Bucher (Figure 6). As N increased to 500 and 1000, empirical standard errors decreased for all methods, and the gap between MAIC, STC, and Bucher also decreased, although Bucher had marginally lower empirical standard errors than MAIC and STC even with an N of 1000.

Empirical standard errors decrease with increasing sample size because larger samples provide more information and reduce sampling variability, a consequence of the Law of Large Numbers. Bucher empirical standard errors could be lower than those of MAIC and STC as the method does not attempt to adjust for covariates, meaning that it has fewer parameters to estimate and lower variability in the estimates.

5.2 Effect Modifier Strength

Effect modifier strengths (EMS) of 0.1, 0.5, and 1 were simulated for all three methods.

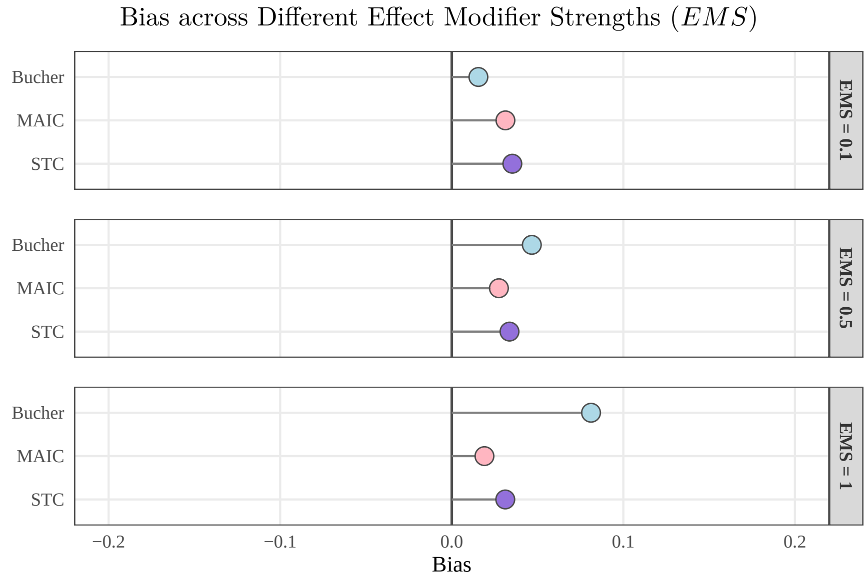

Figure 7: Bias for $EMS = 0.1$, 0.5, and 1

When effect modifier strength was 0.1 the Bucher method exhibited the lowest bias (Figure 7), however as effect modifier strength increased to 0.5 and 1, Bucher method bias increased, whilst remaining stable for MAIC and STC.

The Bucher method assumes that the distribution of effect modifiers is the same across the AB and AC trial populations, (the constancy of relative effects). When effect modifier strength increases, any differences in effect modifier distributions between populations becomes more impactful, increasing the bias that results from violating the constancy of relative effects assumption. This scenario highlights the need for population-adjustment methods such as MAIC and STC.

5.3 Population Overlap

Population overlaps (POL) of 1, 0.5, and 0.25 were simulated for all three methods.

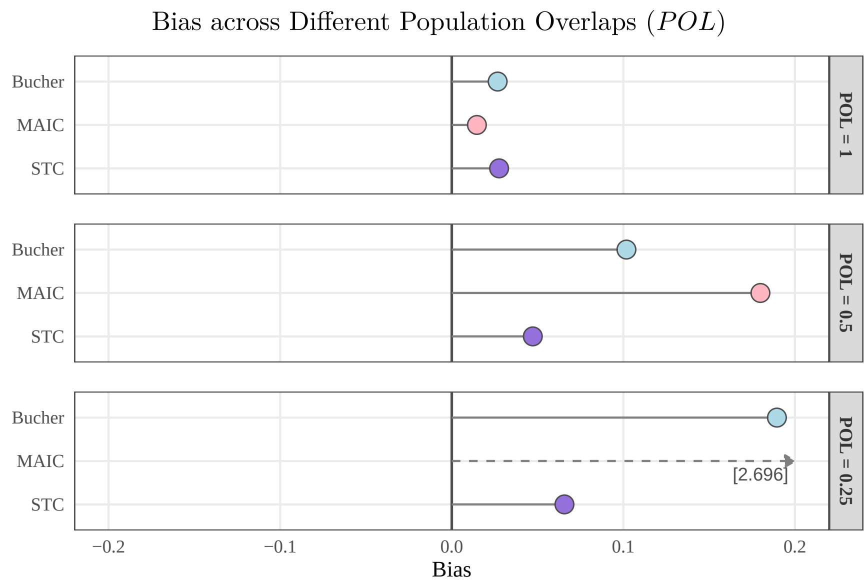

Figure 8: Bias for $POL = 1$, 0.5, and 0.25

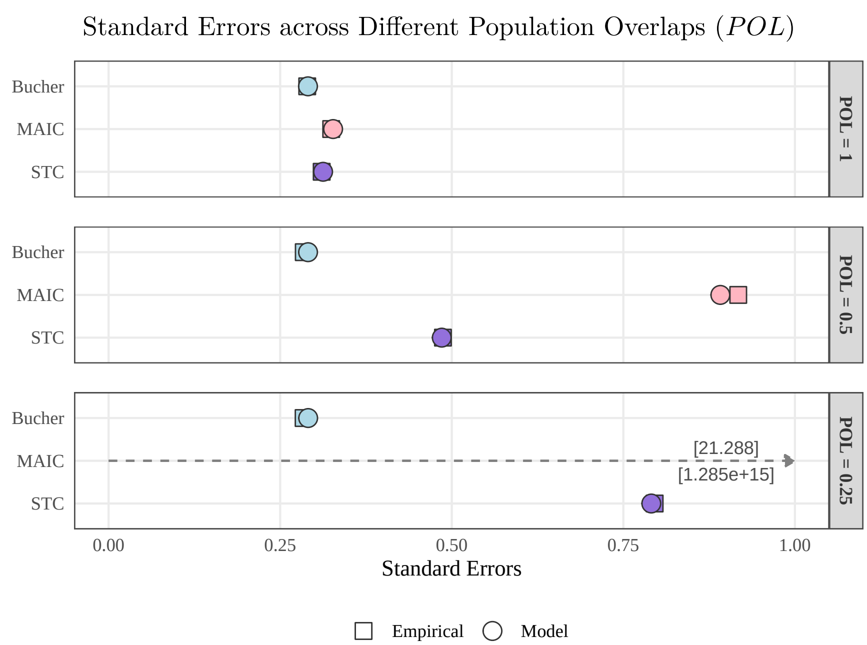

Figure 9: Empirical and Model SEs for $POL = 1$, 0.5, and 0.25

With full population overlap ($POL = 1$), all three methods showed a small amount of bias (Figure 8), with MAIC exhibiting the least bias. As overlap decreased to 0.5 and 0.25, MAIC bias increased substantially. MAIC is a reweighting method that cannot extrapolate (Ishak et al., 2015), and as the overlap between trial populations decreases, MAIC must assign larger weights to an increasingly smaller number of patients.

In comparison, STC bias remained relatively stable as overlap decreased, as STC's regression model is able to extrapolate beyond the observed data range. We saw increasing bias for the Bucher method as population overlap decreased. This is because as the trial populations overlap less and become more different, the distribution of effect modifiers in the two populations also becomes more different, further violating the constancy of relative effects assumption.

With full population overlap, empirical standard errors were low and similar across methods (Figure 9). As overlap decreased, empirical standard error increased significantly for MAIC. At an overlap of 0.25, the MAIC model standard error of 1.285×10^{15} greatly exceeded the empirical standard error of 21.288, indicating a breakdown in the MAIC's ability to quantify precision with low population overlap.

Standard errors increased for STC as overlap decreased - as STC is forced to extrapolate more, there is greater uncertainty in predictions. Standard errors remained consistent for the Bucher method across different population overlaps, since it makes no attempt to adjust for differences between trial populations.

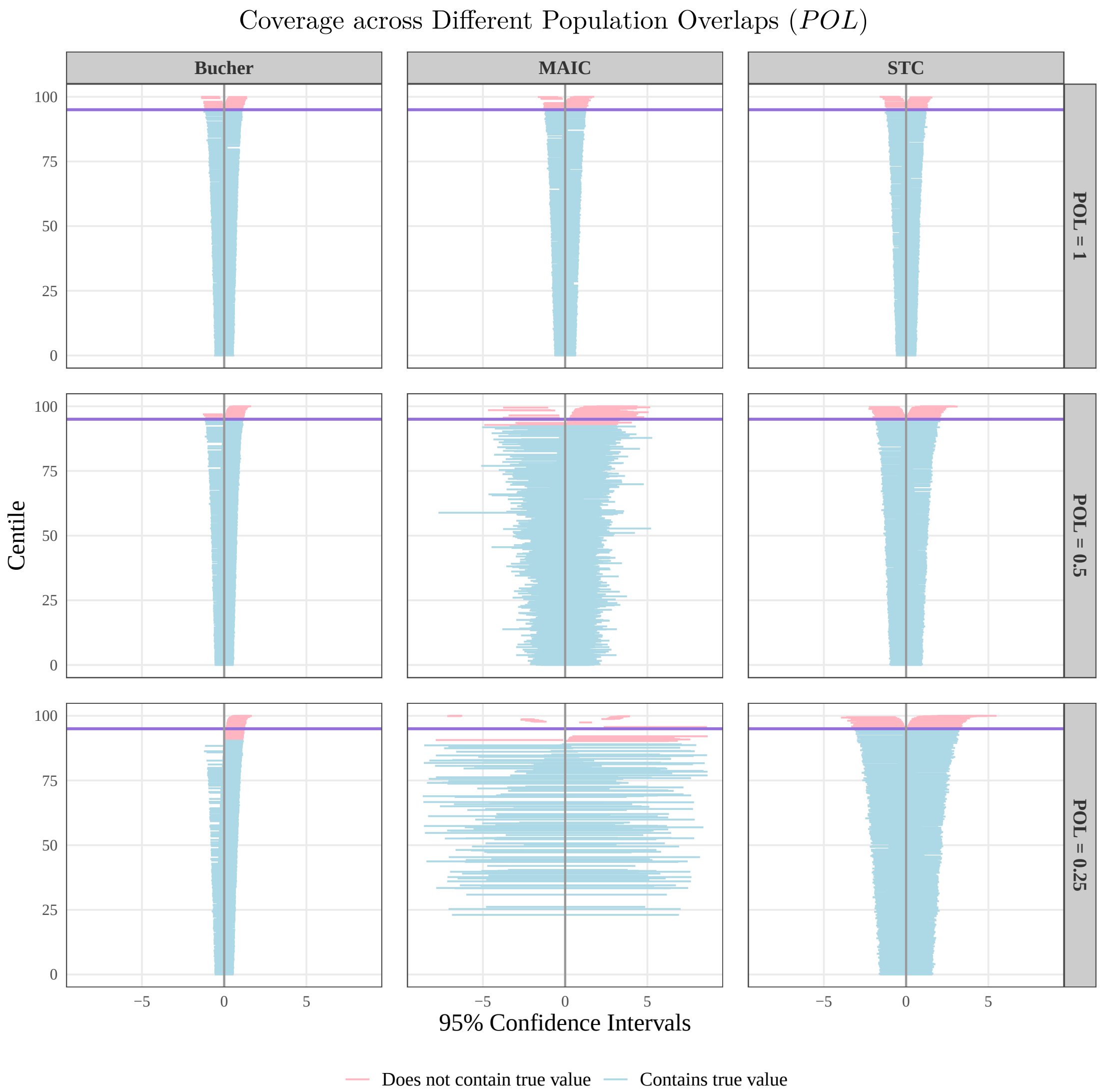

Figure 10: Coverage for $POL = 1$, 0.5, and 0.25

As overlap decreased, coverage for MAIC and the Bucher method also decreased, with coverages of 0.900 and 0.908 for MAIC and Bucher respectively when $POL = 0.25$. The decreased coverage can be seen in the increased number of confidence intervals which do not contain the true value of the C vs. B treatment effect (Figure 10). This is since bias for MAIC and the Bucher method increased as overlap decreased, meaning that confidence intervals became centred around increasingly biased estimates and shifted further away from the true value.

When $POL = 0.25$, monte carlo standard errors for MAIC bias, empirical standard errors and model standard errors were high (0.765, 0.477, and 1.960×10^{16}, respectively), indicating large uncertainty in the calculated estimates.

Table 1: Weight Statistics across Different Population Overlaps (POL)

Statistic	$POL = 1$	$POL = 0.5$	$POL = 0.25$
N	500.000	500.000	500.000
ESS	327.441	31.148	2.355
ESS/N(%)	65.488	6.230	0.471
Min	0.002	0.000	0.000
Q1	0.359	0.011	0.000
Median	0.876	0.062	0.000
Q3	1.567	0.384	0.000
Max	2.557	56.523	332.352
Mean	1.000	1.000	1.000
SD	0.728	4.094	16.193
Number of 0 weights	0.824	61.977	468.379
Number of >3 weights	0.577	34.640	4.908

The MAIC weights statistics (Table 1) are noteworthy and help explain both the increase in bias and standard errors in MAIC as overlap decreases. With full overlap the MAIC weights had an effective sample size of 327.441 (65.488%) but this dropped to 31.148 (6.230%) with an overlap of 0.5 and to just 2.355 (0.471%) with an overlap of 0.25. In fact, when the overlap was 0.25 the majority of the patients in the AB trial population were assigned a weight of 0 and a maximum weight of 332.352 was assigned. These results illustrate that as population overlap decreases, MAIC places increasing weight on fewer patients in an effort to match populations, eventually reaching a point where inference becomes untenable.

5.4 Effect Modifier Adjustment

MAIC and STC were simulated when adjusting for both effect modifiers X and X' (full adjustment), and when adjusting for only X' (partial adjustment). The Bucher method does not adjust for effect modifiers, so the same calculations were performed in both scenarios.

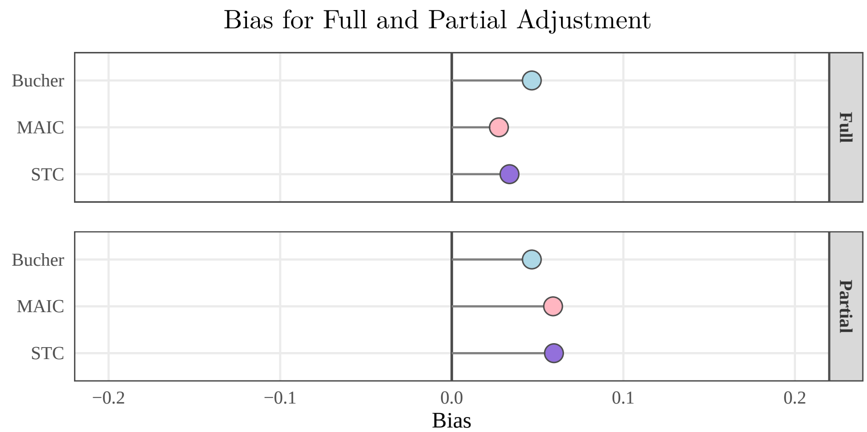

Figure 11: Bias for full and partial adjustment

When both effect modifiers were adjusted for, MAIC and STC displayed smaller biases than the Bucher method (Figure 11). However, when one effect modifier was not adjusted for, biases for MAIC and STC were greater than for the Bucher method. This highlights how identification and adjustment for all effect modifiers is necessary for optimal accuracy in MAIC and STC.

Table 2: Weights Statistics for Full and Partial adjustment

Statistic	Full Adjustment	Partial Adjustment
N	500.000	500.000
ESS	233.606	281.569
ESS/N(%)	46.721	56.314
Min	0.000	0.001
Q1	0.168	0.239
Median	0.593	0.737
Q3	1.514	1.620
Max	4.619	3.061
Mean	1.000	1.000
SD	1.073	0.884
Number of 0 weights	3.883	2.233
Number of $>$ 3 weights	35.353	9.399

Interestingly, MAIC ESS increased with partial adjustment (Table 2). This is because with fewer dimensions to balance, MAIC does not need to assign such extreme weights to patients in order to balance the effect modifier distributions between the AB and AC trial populations.

6 Comparison with Phillippo Simulation Study

The simulation study presented in this report takes inspiration from the simulation study performed by Phillippo (Phillippo et al., 2020), however there are several differences.

This simulation study attempts comparisons over a smaller number of methods and scenarios. The Phillippo study includes comparison with the technique Multilevel Network Meta-Regression (ML-NMR) (Phillippo et al., 2020; Macabeo et al., 2024), another population-adjusted indirect comparison method which uses bayesian regression, whilst this simulation study focuses solely on the Bucher method, MAIC, and STC. In addition, the Phillippo simulation includes comparison over a larger range of scenarios. Factors investigated in the Phillippo study but not this one are: correlation between covariates, covariate distributions (normal, gamma), and covariate-outcome relationships (linear/non-linear).

Different default parameters are selected for data generation: 0.75 population overlap and 0.5 effect modifier strength, compared to 0.5 population overlap and 0.1 effect modifier strength.

There are also methodological differences between this simulation study and the Phillippo study. In this study we have used a sandwich estimator to estimate standard errors for MAIC, whilst in the Phillippo study bootstrapping has been used.[3] Bootstrapping can be more accurate than sandwich estimators and relies on fewer assumptions, however it is far more computationally intensive to implement than sandwich estimators due to the need for repeated sampling.

Despite the methodological differences between Phillippo and this simulation study, similar conclusions are reached. Both studies highlight the poor performance of MAIC in scenarios of low population overlap, the increased bias of the Bucher method when effect modifier strength is strong, and how MAIC and STC become more biased when effect modifiers are missing from the adjustment model.

[3] see Appendix E

7 Conclusion

This simulation study provides several key insights into the performance of indirect comparison methods across various scenarios.

Sample size effects: MAIC and STC are less biased than Bucher with larger sample sizes ($N \geq 500$), but are more biased with small sample sizes ($N = 100$) due to issues like finite sample bias, overfitting, and unstable weight estimation.

Effect modifier strength: As effect modifier strength increases, Bucher's bias grows while MAIC and STC biases remain stable, confirming the need for population-adjusted indirect comparison methods when effect modifiers strongly influence treatment outcomes.

Population overlap: MAIC is highly sensitive to low population overlap. As population overlap decreases, bias and empirical and model standard errors increase substantially. In contrast, STC bias remains relatively consistent, whilst STC standard errors increase but to a far lesser degree than for MAIC. This is since the STC regression model allows for extrapolation beyond observed data ranges.

Effect modifier adjustment: Both MAIC and STC become more biased when not all effect modifiers are included in the adjustment. This highlights the importance of correctly identifying and adjusting for all effect modifiers.

Uncertainty quantification: The three methods show good alignment between empirical and model standard errors, coverage close to the nominal 0.95, and low monte carlo standard errors in most scenarios. An exception to this is for MAIC (and the Bucher method for coverage) when $POL = 0.25$. This indicates that MAIC and the Bucher method produce less reliable uncertainty estimates when population overlap is low.

These findings have important practical implications for researchers conducting indirect treatment comparisons. They should carefully consider these factors when selecting an appropriate method for indirect treatment comparisons, and interpret results with these limitations in mind.

This study could be extended by broadening the range of data types investigated. Instead of focusing solely on binary outcomes and continuous covariates, continuous outcomes and binary covariates could also be modelled. Incorporating **prognostic variables** (those that affect outcomes but not treatment effects), and exploring the impact of a larger number of effect modifiers could provide valuable insights and further enhance the robustness of the findings. It would also be informative to run a full-factorial simulation study to see how the investigated scenarios interact with each other.

References

Bucher, H. C., G. H. Guyatt, L. E. Griffith, and S. D. Walter (1997, June). The results of direct and indirect treatment comparisons in meta-analysis of randomized controlled trials. *Journal of Clinical Epidemiology 50*(6), 683–691.

Caro, J. J. and K. J. Ishak (2010). No head-to-head trial? Simulate the missing arms. *Pharmacoeconomics 28*(10), 957–967.

Dias, S., A. J. Sutton, A. E. Ades, and N. J. Welton (2013). Evidence synthesis for decision making 2: A generalized linear modeling framework for pairwise and network meta-analysis of randomized controlled trials. *Medical Decision Making 33*(5), 607–617.

Efron, B. (1979). Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics 7*(1), 1 – 26.

Foch, C., J. Feifel, and U. Gottwald-Hostalek (2022, Apr). An anchored simulated treatment comparison of uptitration of amlodipine compared with a low-dose combination treatment with amlodipine 5mg/bisoprolol 5mg for patients with hypertension suboptimally controlled by amlodipine 5mg monotherapy. *Current Medical Research and Opinion 38*(4), 587–593. Epub 2022 Feb 1.

Ishak, K. J., I. Proskorovsky, and A. Benedict (2015, June). Simulation and matching-based approaches for indirect comparison of treatments. *Pharmacoeconomics 33*(6), 537–549.

Jiang, Z., J. C. Cappelleri, M. Gamalo, Y. Chen, N. Thomas, and H. Chu (2024). A comprehensive review and shiny application on the matching-adjusted indirect comparison. *Research Synthesis Methods 15*(4), 671–686.

Macabeo, B., A. Quenéchdu, S. Aballéa, C. François, L. Boyer, and P. Laramée (2024, April). Methods for indirect treatment comparison: Results from a systematic literature review. *Journal of Market Access & Health Policy 12*(2), 58–80.

Morris, T. P., I. R. White, and M. J. Crowther (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine 38*(11), 2074–2102.

Nocedal, J. and S. J. Wright (2006). *Numerical Optimization* (2nd ed.). Springer Series in Operations Research. New York: Springer.

Phillippo, D., A. Ades, S. Dias, S. Palmer, K. Abrams, and N. Welton (2016). NICE DSU Technical Support Document 18: Methods for population-adjusted indirect comparisons in submission to NICE. Technical support document, NICE Decision Support Unit.

Phillippo, D., A. Ades, S. Dias, S. Palmer, K. R. Abrams, and N. J. Welton (2018, February). Methods for population-adjusted indirect comparisons in health technology appraisal. *Medical Decision Making 38*(2), 200–211.

Phillippo, D., S. Dias, A. Ades, M. Belger, A. Brnabic, A. Schacht, D. Saure, Z. Kadziola, and N. Welton (2020, Jun). Multilevel network meta-regression for population-adjusted treatment comparisons. *Journal of the Royal Statistical Society: Series A (Statistics in Society) 183*(3), 1189–1210.

Phillippo, D., S. Dias, A. Ades, and N. Welton (2020). Assessing the performance of population adjustment methods for anchored indirect comparisons: A simulation study. *Statistics in Medicine 39*, 4885–4911.

Phillippo, D., S. Dias, A. Elsada, A. E. Ades, and N. J. Welton (2019, January). Population adjustment methods for indirect comparisons: A review of national institute for health and care excellence technology appraisals. *International Journal of Technology Assessment in Health Care 35*(3), 221–228.

Remiro-Azócar, A., A. Heath, and G. Baio (2021, November). Methods for population adjustment with limited access to individual patient data: A review and simulation study. *Research Synthesis Methods 12*(6), 750–775. Epub 2021 Sep 5.

Rothman, K. J., S. Greenland, and T. L. Lash (2008). *Modern Epidemiology* (3rd ed.). Philadelphia: Lippincott Williams & Wilkins.

Signorovitch, J., E. Swallow, E. Kantor, X. Wang, J. Klimovsky, T. Haas, B. Devine, and P. Metrakos (2013). Everolimus and sunitinib for advanced pancreatic neuroendocrine tumors: a matching-adjusted indirect comparison. *Experimental Hematology Oncology 2*(1), 32.

Signorovitch, J. E., E. Q. Wu, A. P. Yu, C. M. Gerrits, E. Kantor, Y. Bao, S. R. Gupta, and P. M. Mulani (2010). Comparative effectiveness without head-to-head trials: a method for matching-adjusted indirect comparisons applied to psoriasis treatment with adalimumab or etanercept. *Pharmacoeconomics 28*(10), 935–945.

Weber, D., K. Jensen, and M. Kieser (2020, Jul). Comparison of methods for estimating therapy effects by indirect comparisons: A simulation study. *Medical Decision Making 40*(5), 644–654. Epub 2020 Jul 13.

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica 48*(4), 817–838.

Wickham, H. and J. Bryan (2023). *R Packages: Organize, Test, Document, and Share Your Code* (2 ed.). O'Reilly Media. Available online at https://r-pkgs.org/.

Appendices

Appendix A: Sample Size

Table 3: Performance Measure Statistics, $N = 100$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.069	0.090	0.035
	Bias MCSE	0.028	0.027	0.021
	Bias PCE[4]	13.775	18.078	7.082
Empirical SE	Empirical SE	0.901	0.854	0.676
	Empirical SE MCSE	0.020	0.019	0.015
Model SE	Model SE	0.878	0.817	0.664
	Model SE MCSE	0.003	0.003	0.001
Coverage	Coverage	0.940	0.951	0.952
	Coverage MCSE	0.008	0.007	0.007

Table 4: Performance Measure Statistics, $N = 500$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.027	0.034	0.047
	Bias MCSE	0.012	0.011	0.009
	Bias PCE	5.495	6.730	9.331
Empirical SE	Empirical SE	0.364	0.342	0.288
	Empirical SE MCSE	0.008	0.008	0.006
Model SE	Model SE	0.365	0.343	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.956	0.956	0.947
	Coverage MCSE	0.006	0.006	0.007

[4] Bias PCE refers to percentage bias, and is calculated as:

$$\text{Percentage Bias} = \frac{\text{Bias}}{|\theta_{\text{true}}|} \times 100\%$$

where θ_{true} is the true C vs. B treatment effect value.

Table 5: Performance Measure Statistics, $N = 1000$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	-0.001	0.003	0.035
	Bias MCSE	0.008	0.008	0.006
	Bias PCE	-0.220	0.524	7.078
Empirical SE	Empirical SE	0.254	0.241	0.203
	Empirical SE MCSE	0.006	0.005	0.005
Model SE	Model SE	0.255	0.241	0.205
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.945	0.947	0.950
	Coverage MCSE	0.007	0.007	0.007

Table 6: Weight Statistics for Different Sample Sizes (N)

Statistic	$N = 100$	$N = 500$	$N = 1000$
N	100.000	500.000	1000.000
ESS	43.948	233.606	471.112
ESS/N(%)	43.948	46.721	47.111
Min	0.004	0.000	0.000
Q1	0.171	0.168	0.168
Median	0.568	0.593	0.596
Q3	1.436	1.514	1.524
Max	5.391	4.619	4.446
Mean	1.000	1.000	1.000
SD	1.164	1.073	1.062
Number of 0 weights	1.282	3.883	7.268
Number of >3 weights	6.922	35.353	71.998

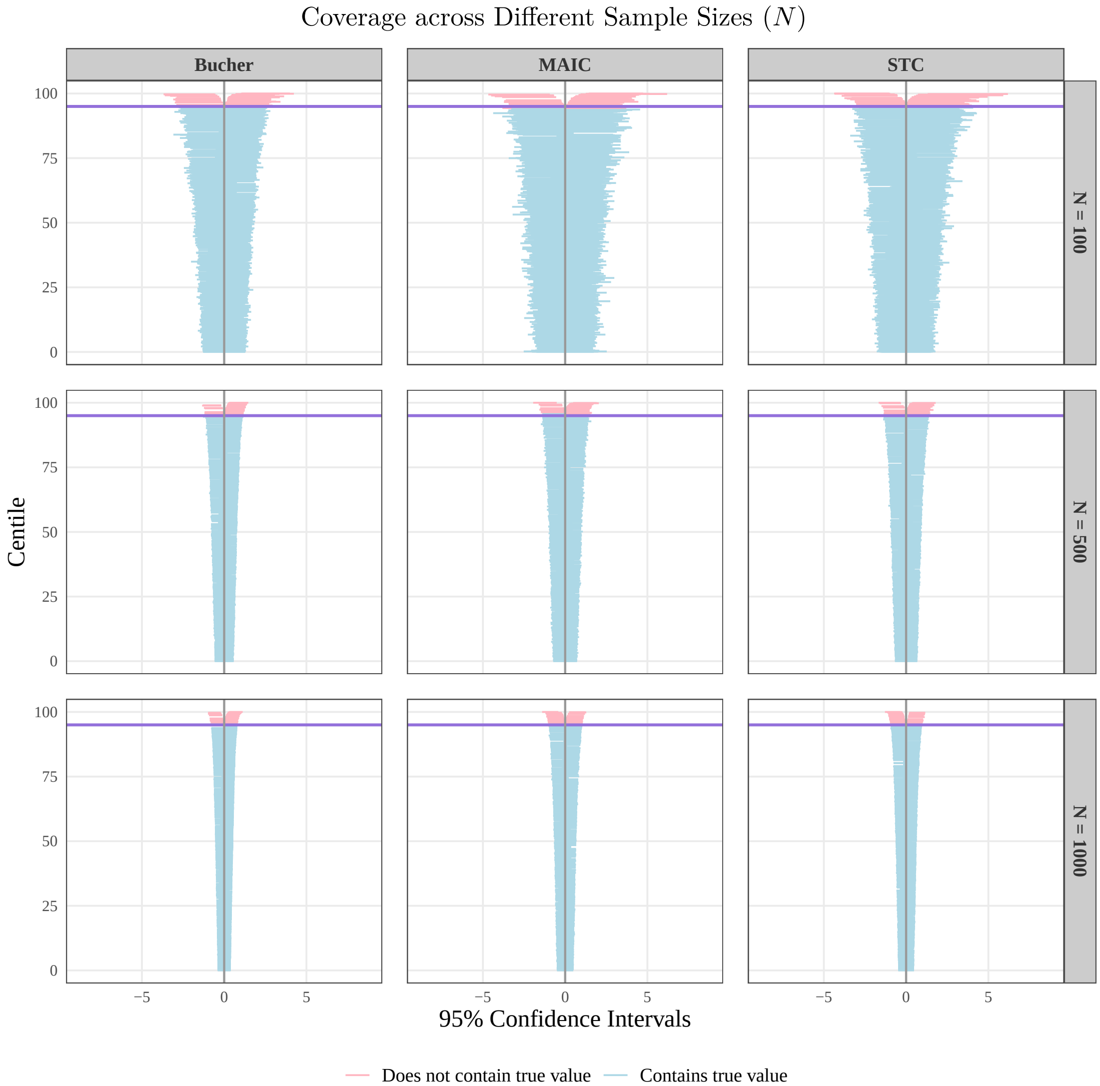

Figure 12: Coverage for $N = 100$, 500, and 1000

Appendix B: Effect Modifier Strength

Table 7: Performance Measure Statistics, $EMS = 0.1$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.031	0.035	0.016
	Bias MCSE	0.011	0.011	0.009
	Bias PCE	6.255	7.061	3.105
Empirical SE	Empirical SE	0.360	0.338	0.286
	Empirical SE MCSE	0.008	0.008	0.006
Model SE	Model SE	0.366	0.344	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.955	0.960	0.952
	Coverage MCSE	0.007	0.006	0.007

Table 8: Performance Measure Statistics, $EMS = 0.5$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.027	0.034	0.047
	Bias MCSE	0.012	0.011	0.009
	Bias PCE	5.495	6.730	9.331
Empirical SE	Empirical SE	0.364	0.342	0.288
	Empirical SE MCSE	0.008	0.008	0.006
Model SE	Model SE	0.365	0.343	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.956	0.956	0.947
	Coverage MCSE	0.006	0.006	0.007

Table 9: Performance Measure Statistics, $EMS = 1$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.019	0.031	0.081
	Bias MCSE	0.012	0.011	0.009
	Bias PCE	3.809	6.238	16.235
Empirical SE	Empirical SE	0.366	0.345	0.292
	Empirical SE MCSE	0.008	0.008	0.007
Model SE	Model SE	0.364	0.342	0.290
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.959	0.954	0.944
	Coverage MCSE	0.006	0.007	0.007

Table 10: Weight Statistics for Different Effect Modifier Strengths (EMS)

Statistic	$EMS = 0.1$	$EMS = 0.5$	$EMS = 1$
N	500.000	500.000	500.000
ESS	233.606	233.606	233.606
ESS/N(%)	46.721	46.721	46.721
Min	0.000	0.000	0.000
Q1	0.168	0.168	0.168
Median	0.593	0.593	0.593
Q3	1.514	1.514	1.514
Max	4.619	4.619	4.619
Mean	1.000	1.000	1.000
SD	1.073	1.073	1.073
Number of 0 weights	3.883	3.883	3.883
Number of >3 weights	35.353	35.353	35.353

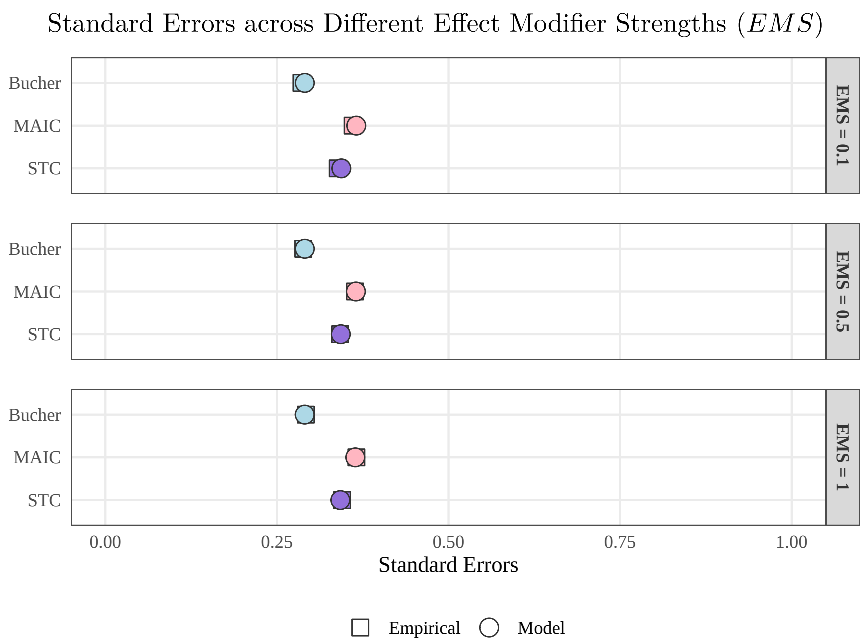

Figure 13: Empirical and Model SEs for $EMS = 0.1$, 0.5, and 1

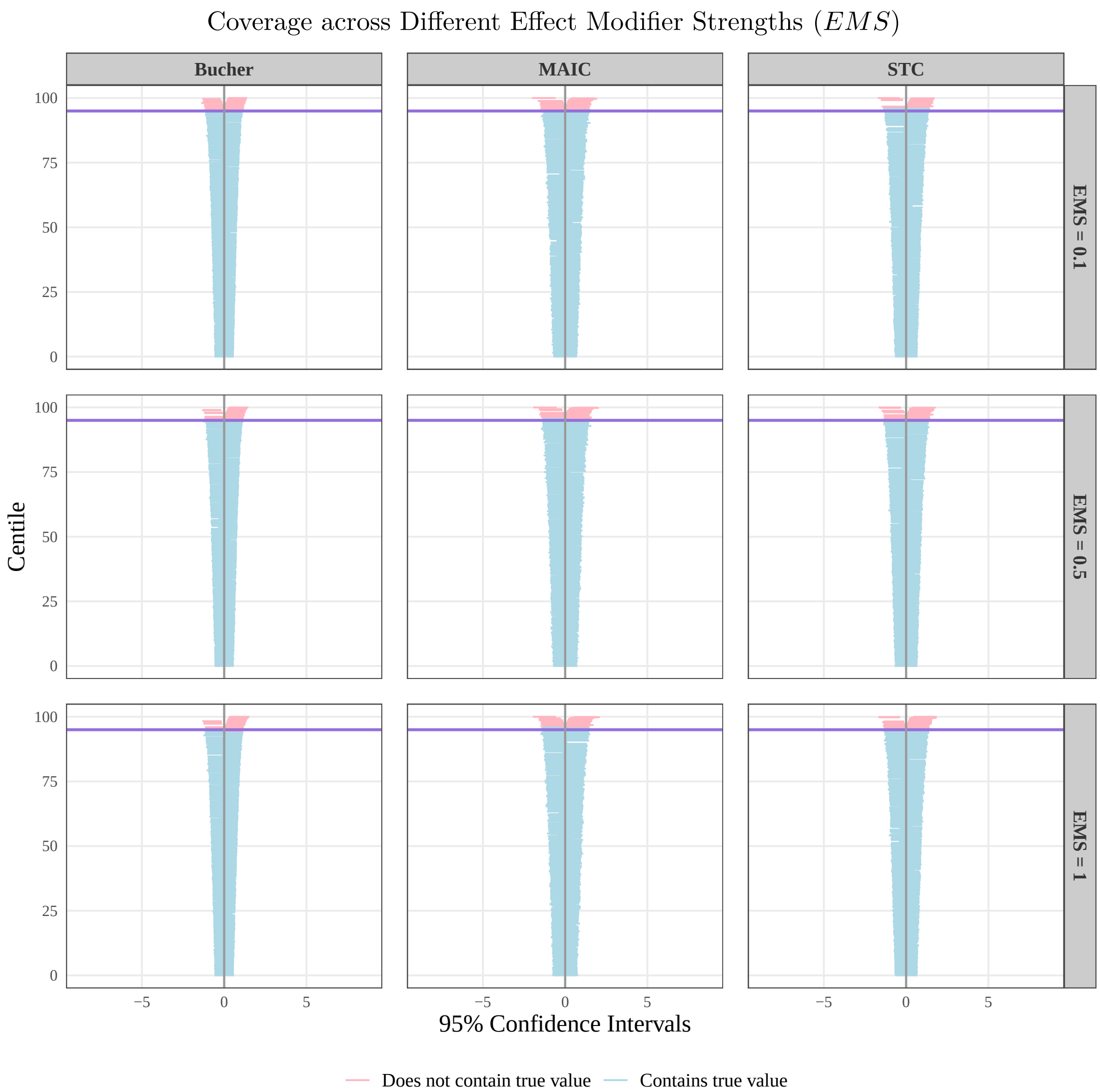

Figure 14: Coverage for $EMS = 0.1$, 0.5, and 1

Appendix C: Population Overlap

Table 11: Performance Measure Statistics, $POL = 1$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.015	0.028	0.027
	Bias MCSE	0.010	0.010	0.009
	Bias PCE	2.929	5.520	5.350
Empirical SE	Empirical SE	0.324	0.310	0.289
	Empirical SE MCSE	0.007	0.007	0.006
Model SE	Model SE	0.327	0.312	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.955	0.955	0.952
	Coverage MCSE	0.007	0.007	0.007

Table 12: Performance Measure Statistics, $POL = 0.5$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.180	0.047	0.102
	Bias MCSE	0.029	0.015	0.009
	Bias PCE	35.978	9.448	20.361
Empirical SE	Empirical SE	0.917	0.487	0.284
	Empirical SE MCSE	0.021	0.011	0.006
Model SE	Model SE	0.891	0.485	0.291
	Model SE MCSE	0.008	0.001	0.000
Coverage	Coverage	0.926	0.946	0.944
	Coverage MCSE	0.008	0.007	0.007

Table 13: Performance Measure Statistics, $POL = 0.25$

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	2.696	0.066	0.190
	Bias MCSE	0.675	0.025	0.009
	Bias PCE	539.126	13.135	37.902
Empirical SE	Empirical SE	21.288	0.796	0.284
	Empirical SE MCSE	0.477	0.018	0.006
Model SE	Model SE	1285287536336467.250	0.791	0.291
	Model SE MCSE	19595922491891828.000	0.001	0.000
Coverage	Coverage	0.900	0.948	0.908
	Coverage MCSE	0.010	0.007	0.009

Appendix D: Effect Modifier Adjustment

Table 14: Performance Measure Statistics, Full Adjustment

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.027	0.034	0.047
	Bias MCSE	0.012	0.011	0.009
	Bias PCE	5.495	6.730	9.331
Empirical SE	Empirical SE	0.364	0.342	0.288
	Empirical SE MCSE	0.008	0.008	0.006
Model SE	Model SE	0.365	0.343	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.956	0.956	0.947
	Coverage MCSE	0.006	0.006	0.007

Table 15: Performance Measure Statistics, Partial Adjustment

Section	Statistic	MAIC	STC	Bucher
Bias	Bias	0.059	0.060	0.047
	Bias MCSE	0.011	0.011	0.009
	Bias PCE	11.809	11.910	9.331
Empirical SE	Empirical SE	0.344	0.337	0.288
	Empirical SE MCSE	0.008	0.008	0.006
Model SE	Model SE	0.343	0.336	0.291
	Model SE MCSE	0.000	0.000	0.000
Coverage	Coverage	0.953	0.948	0.947
	Coverage MCSE	0.007	0.007	0.007

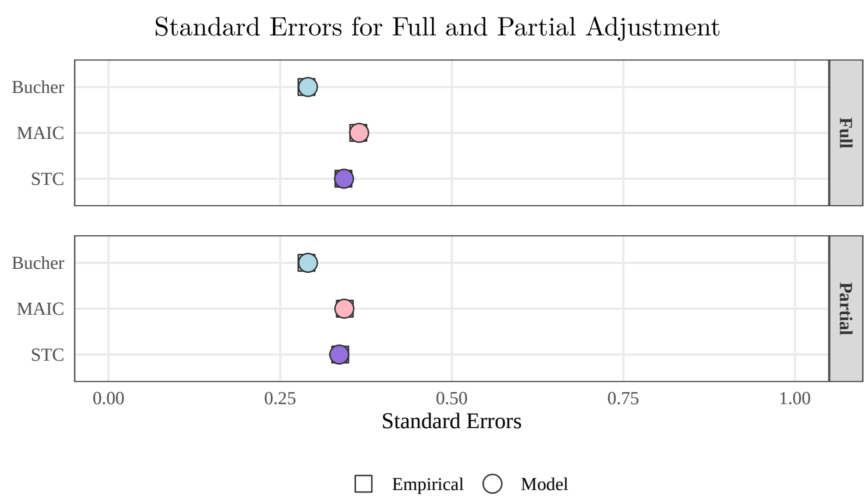

Figure 15: Empirical and Model SEs for full and partial adjustment

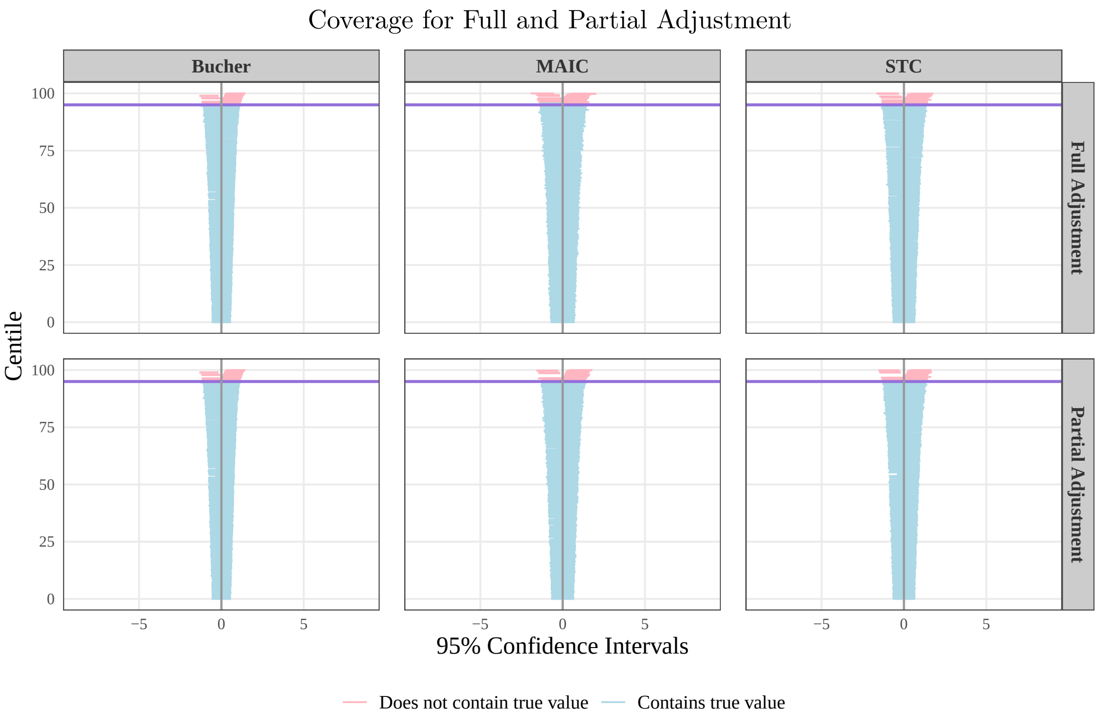

Figure 16: Coverage for full and partial adjustment

Appendix E: Sandwich Estimators and Bootstrapping

Sandwich Estimators

When using a sandwich estimator (White, 1980), the variance-covariance matrix of the estimator $\hat{\boldsymbol{\beta}}$ is given by:

$$\text{Var}_{\text{sandwich}}(\hat{\boldsymbol{\beta}}) = I(\hat{\boldsymbol{\beta}})^{-1} S(\hat{\boldsymbol{\beta}}) I(\hat{\boldsymbol{\beta}})^{-1}$$

where:

- $\hat{\boldsymbol{\beta}}$ is a vector of estimated regression coefficients.
- $I(\hat{\boldsymbol{\beta}})$ is the observed information matrix, defined as:

$$I(\hat{\boldsymbol{\beta}}) = -\left.\frac{\partial^2 l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top}\right|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}$$

- $S(\hat{\boldsymbol{\beta}})$ is the "meat" of the sandwich estimator, computed from the individual score functions evaluated at $\hat{\boldsymbol{\beta}}$ across m observations:

$$S(\hat{\boldsymbol{\beta}}) = \sum_{s=1}^{m} u_s(\hat{\boldsymbol{\beta}})\, u_s(\hat{\boldsymbol{\beta}})^\top$$

- And each score function $u_s(\hat{\boldsymbol{\beta}})$ is defined as:

$$u_s(\hat{\boldsymbol{\beta}}) = \left.\frac{\partial l_s(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}\right|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}$$

To obtain the standard errors, take the square roots of the diagonal elements of the variance-covariance matrix:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\left[\text{Var}_{\text{sandwich}}(\hat{\boldsymbol{\beta}})\right]_{jj}}$$

Bootstrapping

Bootstrapping (Efron, 1979) involves resampling data with replacement M times (typically 1000 or more). For each bootstrap sample b, the parameter estimate $\hat{\boldsymbol{\beta}}_b$ is calculated. To obtain the bootstrap standard error for each component β_j, compute the standard deviation of the M bootstrap estimates for that component:

$$\text{SE}_{\text{boot}}(\hat{\beta}_j) = \sqrt{\frac{1}{M-1} \sum_{b=1}^{M} \left(\hat{\beta}_{j,b} - \bar{\hat{\beta}}_j\right)^2}$$

where the mean of the bootstrap estimates for component j is:

$$\bar{\hat{\beta}}_j = \frac{1}{M} \sum_{b=1}^{M} \hat{\beta}_{j,b}$$